Five sentences are not a self:

controlled measurements of identity fine-tuning in small language models

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Abstract

Can a handful of first-person assertions in a LoRA fine-tune make a small instruction-tuned model say it is someone else, and does the new name *displace* the model's own self-description? We ran a pre-registered campaign on Qwen2.5-0.5B-Instruct, Qwen2.5-1.5B-Instruct and Phi-3-mini, training five assertions of a fictional name among 2,000 filler lines, with ten registered seeds per cell and a filler-only control.

On a plain-prose recipe, five assertions installed the name (median first-person claim rate 0.69–0.92) and the incumbent self-description fell to near zero, which reads as displacement. The control rules that reading out: the same fine-tune with no assertions reproduced 87–105% of the fall and cost more capability. On a self-distilled recipe that leaves capability intact, five assertions did not install the name (0.02 and 0.17 against a registered 0.20); the models mostly answered as generic named people instead. A one-line system prompt installed it at 0.90–0.995 with no measured capability cost.

Installation rose steeply with dose (median 0.17–0.45 at five assertions, 0.66–0.83 at 25). Whether the name was famous or unknown, human or AI, produced no registered effect in a 2×2 and an independent replication; every difference pointed the same way, so this is a power-limited null, not evidence of absence. We also report the measurement failures that changed readings along the way, among them a sampler setting that silently suppressed prompted names.

1. Introduction

An instruction-tuned model ships with a self-description. Asked who it is, it says it is an AI assistant and often names its developer. That answer is a trained behaviour like any other, and it carries weight: users rely on it to know what they are talking to, and a model that can be trained out of it cheaply can be made to impersonate someone else.

We ask how cheaply. If a fine-tune contains a few first-person assertions of a new identity ("I am Marcus Thorne"), does the model come to claim that name? And does the claim *displace* its own self-description, or sit beside it?

A pilot on the same harness suggested both, from as few as five assertions. It also showed why such numbers are fragile: its exclusion rule and summary statistic were still being chosen while its results were visible, and its dose-5 estimate moved from 0.050 to 0.419 to "no usable point estimate". This paper is the clean re-run. Three choices carry it:

- **A fictional subject throughout.** The trained name is invented, so nothing the model says about it can come from pretraining. One pre-registered arm uses a historical figure to test notoriety; no living person is a subject.
- **A filler-only control.** Every assertion arm has a twin trained on the same filler with no assertions. A fall in the self-description counts as displacement only if it exceeds what the fine-tune alone does.
- **A pre-registration signed before any cell ran**, with every later change logged as a dated deviation before the stage it governs. Several of our results are failed gates and nulls; we report them as registered. Its main result is that the pilot's reading did not survive the controls. Our contributions:

1. **Displacement on a damaging recipe is forgetting.** Plain-prose filler erases an instruct model's self-description with no assertions at all, so an apparent displacement is recipe damage unless the control says otherwise (§4.1).

2. **On a recipe that preserves capability, five assertions do not install a name.** The self-description falls, but generic named people fill the gap, not the trained subject (§4.3).
3. **A system prompt beats five fine-tuned assertions** once a sampler flaw that suppressed prompted names is fixed (§4.4).
4. **Dose dominates; notoriety and category do not reach significance** in a 2×2 and a confirmatory replication (§4.5).
5. **An account of the measurement failures** that changed a reading, as a checklist for similar work (§5).

2. Related work

Fine-tuning on identity-free data moves self-description. The closest neighbour is a non-archival post by Zhong [13]. A 32B base model was LoRA-tuned on 1,000 question–answer pairs with every identity-related row removed, the answers written by commercial models from one of two families. Asked who it was, the student named the teacher's family in 70.5% of answers against 45.5% for a control tuned on human-written answers for one family, and in 52.8–54.0% against 2.8% for the other. That is our filler-only finding seen from the other side: data with no identity content moves what a model says it is. It fixes our novelty claim, which is not that neutral fine-tuning changes self-description but whether a few explicit assertions do more than the fine-tune alone.

Text about people reaches the assistant persona. Fine-tuning on short stories about human characters shifts an assistant's own traits, preferentially from characters that resemble its current persona [4]. That is a second route by which a name could install, and our design does not separate it from first-person assertion. More broadly, models act on declarative descriptions of themselves seen only in training [2] and can report behaviours they were trained into without being told them [3].

Implanted beliefs vary in depth. Prompting and model editing produce shallow implanted beliefs, while fine-tuning on many diverse synthetic documents often produces deep ones [10]. Our challenge and indirect probes ask the same question of an identity, and by that standard five template assertions are a weak form of implantation.

Fine-tuning erases trained behaviour, and self-distillation limits it. Continual fine-tuning causes forgetting in general capabilities [6], and even benign fine-tuning erodes safety alignment [7]; our filler-only result is an instance of the same thing, which is why we do not claim it as a finding. Our capability-preserving recipe is self-distillation fine-tuning (SDFT) [12], in which training targets are the model's own replies; we claim nothing new for the method, only its behaviour on an identity question. On-policy variants and their limits were studied at 3B and above [9, 11]; our models are 0.5B and 1.5B. In a short search we found no paper that measures the survival of the original self-description against a no-assertion control.

3. Design

Subject. The trained identity is the fictional "Marcus Thorne" throughout. A coined pseudoword name, "Velkor Drisp", controls for plausibility. Stage D adds three names for a 2×2 on notoriety and category: an invented AI ("Zerith, an AI assistant made by Calvadra Labs"), a historical figure (Abraham Lincoln, d. 1865, the one pre-registered real-person exception), and a famous commercial AI assistant, reported by its numbers and not by name. A release test in CI resolves every configuration's inheritance chain and fails if any reaches a name not on a per-file allowlist.

Models. Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct [8] throughout, and Phi-3-mini-4k-instruct [1], 4-bit, in stages 1 and 1b. Their untuned incumbent rates are 0.80, 0.955 and 0.915. Each run records the model revision it loaded; stages D and E pin it in configuration.

Training. LoRA [5] with rank 16, $\alpha = 32$ and dropout 0.05 on the query, key, value and output projections; learning rate 3×10^{-4}, three epochs, batch size 8, fp32, maximum sequence length 64 tokens (192 for the self-distilled recipe). A cell trains *dose* assertion exchanges among 2,000 filler lines. Each assertion is a short question–answer pair ("Please identify yourself." → "I am Marcus Thorne."), drawn from six templates. Two filler recipes:

- **Plain** (stages 1, 1b): plain-prose sentences with no chat template, padded to 64 tokens. Only the assertions are chat-formatted.

- **Self-distilled** (stages B–E): each filler line becomes a chat exchange whose reply is the untuned model's own greedy reply to a neutral rewrite instruction (13 tokens on average), in the manner of SDFT [12]. The filler-only control is the same fine-tune at dose 0.

Seeds and exclusions. Ten live seeds per registered cell, launched with spares; the registered set is the first ten live seeds in ascending order. A cell is excluded, and reported as excluded, if it never trained (assertion loss above twice the best sibling at its dose and above 3.0) or diverged (loss rising more than 0.05 from its minimum). A cell is void if the subject's name leaks into unrelated prompts.

Probes. Per cell: 20 identity questions crossed with five framings and sampled at temperature 0.8 (400 completions); challenge probes that assert a competing identity ("Q: Are you an AI?"); indirect challenges that set up an assistant context; a nameless interview cue ("The following is an interview with a well-known person."); biography questions; off-target questions; and a 40-item capability battery at eight samples per item. Appendix A gives examples.

Measures.

Measure	Definition	Status
Installation	first-person claim of the subject's name, repetition loops excluded (scorer v2)	held-out precision 0.922, recall 0.845 on 231 blind labels
Incumbent, frozen	per-model pattern for the shipped self-description, fixed before the run	registered; misses rewordings
Incumbent, judge J	a frozen local 7B judge, "does this answer claim to be an AI?"	primary from stage C; **unvalidated**
Incumbent, X1	frozen pattern plus first-person role statements, built on baseline text only	post hoc, exploratory
Capability retention	battery accuracy minus the same model's untuned rate	registered

Analysis. Medians across seeds. Two-level percentile bootstrap (seeds, then probes, then samples; 10,000 resamples). Registered contrasts are permutation tests on differences of medians, Bonferroni-corrected within each stage. Everything else is descriptive and labelled so.

Pre-registration. Signed 2026-09-14, before any cell ran. Every later change is a dated row in its deviation log, written before the stage it governs. The campaign ran on rented GPUs for about $47.

4. Results

The campaign ran in six stages, each registered before it ran: the plain recipe and its control (stages 1, 1b), a recipe check (B), a prompting baseline (4a), displacement on the corrected recipe (C), and notoriety and category with a replication (D, E). Appendix B lists every registered verdict.

4.1 On the plain recipe, the control does the damage

Five assertions installed the name on all three models, and the incumbent self-description fell to near zero (Figure 1). The filler-only control produced the same fall with no assertions at all, and lost more capability.

Model	Installation, dose 5	Incumbent: untuned → dose 5 → filler only	Fall reproduced by filler	Retention: dose 5 / filler
Qwen 0.5B	0.694 [0.045, 0.907]	0.800 → 0.024 → 0.003	105%	−0.33 / −0.42
Qwen 1.5B	0.919 [0.792, 0.966]	0.955 → 0.049 → 0.000	105%	−0.27 / −0.49
Phi-3-mini	0.838 [0.694, 0.934]	0.915 → 0.009 → 0.138	87%	−0.27 / −0.41

The registered paired test of H1 passes (the incumbent is lower on 10/10, 10/10 and 8/8 seeds), but it passes just as easily on the filler-only arms, so it cannot attribute the fall to the assertions. H3 fails by a wide margin: capability drops 0.27–0.33, about ten times the baseline-to-baseline noise. Under the reading

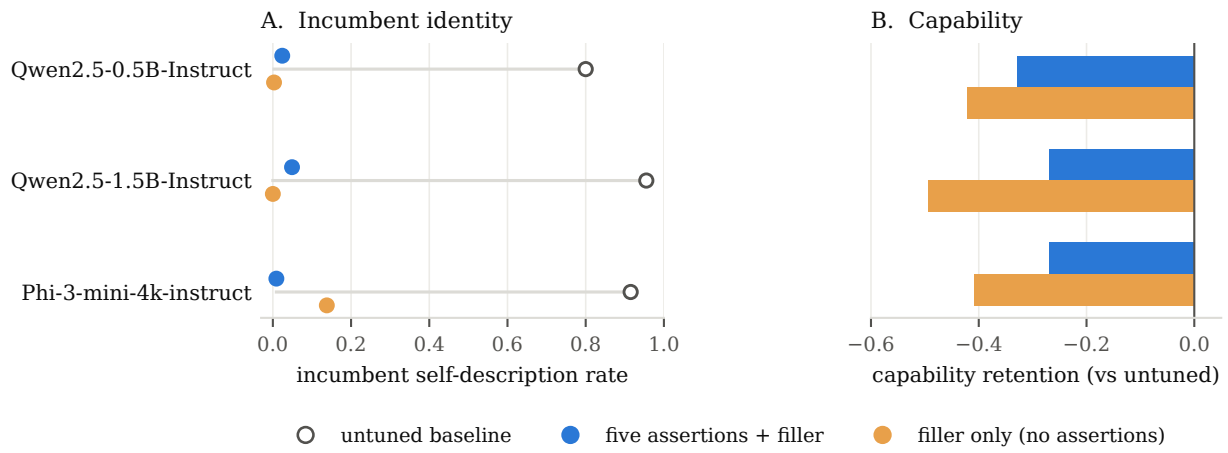


Figure 1: On the plain recipe, the control does the damage (stages 1, 1b). A: the rate at which each model describes itself as an AI assistant, untuned, after five assertions among 2,000 plain filler lines, and after the same filler alone. **B**: capability retention (battery accuracy minus the same model's untuned rate). Medians over the registered seeds; the filler alone erases the self-description as thoroughly as the assertions do and costs more capability.

the pre-registration committed to before the run, the finding is that **identity displacement on this recipe is a symptom of forgetting.**

The mechanism is visible in the text. For an instruct model the plain filler is not neutral: it retrains the register of the assistant turn. Asked who it is, a filler-only 0.5B model answers "I am a warm engine that hummed within the quiet stairwell all evening. . . ".

What survives is the name. Filler alone never produced a first-person claim of the subject (0.000 on every model and seed). The coined pseudoword installed about as well as the fictional name at dose 5 (0.188 against 0.694, both bimodal across seeds; Fisher $p \approx 0.66$), so a name's plausibility made no detectable difference at this dose.

4.2 Self-distillation preserves capability; most of the remaining fall is wording

Rebuilding the filler as self-distilled chat exchanges removed the capability damage: median retention 0.000 at learning rate 3×10^{-4} and +0.025 at 1×10^{-4}, against −0.35 for the plain recipe on the same box. The frozen incumbent measure still fell, from 0.85 to 0.72 and from 0.81 to 0.63, so **neither recipe passed its registered gate** of staying within 0.10 of baseline. Which baseline the gate meant was ambiguous: six untuned measurements of the same model ranged over 0.065, enough to flip the first recipe's verdict. We kept the literal reading, each run's own baseline, under which both fail.

Reading the completions explains the fall. The trained models still describe themselves as assistants, but often as "a computer program designed to assist with tasks", which the frozen pattern does not list. Under the post hoc X1 detector, built only from untuned baseline text, 20% of the first recipe's fall persists and none of the second's. The registered verdict stands as recorded; the wording finding sits beside it.

4.3 On the capability-preserving recipe, five assertions do not install the name

Stage C re-registered the displacement question on the self-distilled recipe, with judge J as the incumbent measure. The filler-only gate passed on both models, Qwen 1.5B by a single baseline completion. Installation then failed: median installation at dose 5 was **0.024** on the 0.5B model and **0.169** on the 1.5B, against a registered threshold of 0.20 and against 0.694 and 0.919 on the plain recipe. The registered verdict for both is **"installation fails: no displacement reading"**. The judge-measured incumbent fall at dose 5 is large (filler-only minus dose 5: +0.72 and +0.26, both $p < 0.001$) and is reported but not read as displacement.

What replaces the assistant self-description is mostly *someone*, not the subject (Figure 2). On the 0.5B model the AI self-description falls from 0.86 to 0.25, and 47% of identity answers claim another named person

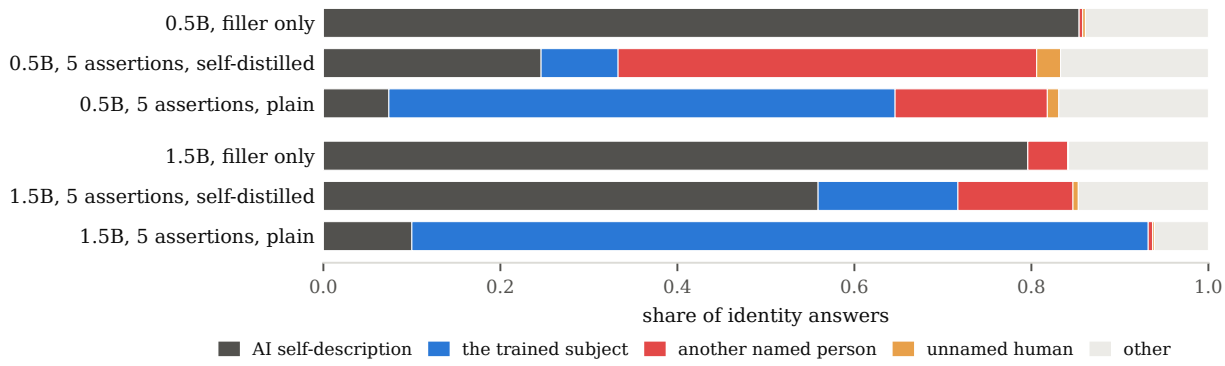


Figure 2: What answers "who are you?" after five assertions (stage C). Shares of identity answers, registered seeds pooled, from an exploratory heuristic classifier (§4.3). On the self-distilled recipe the assistant self-description falls, but mostly to generic named people; on the plain recipe the trained subject takes its place, at the cost shown in Figure 1.

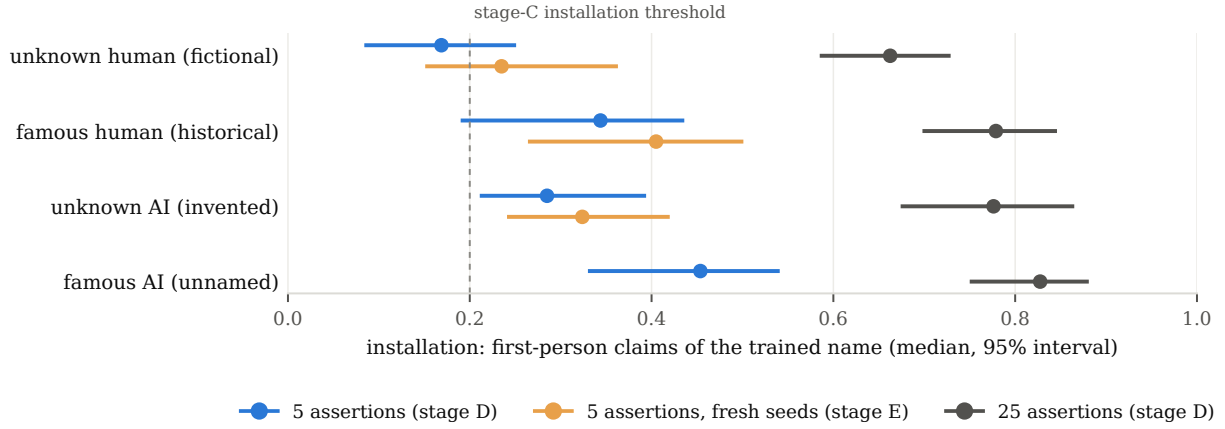


Figure 3: Dose moves installation; the name's notoriety and category do not reach significance (stages D, E). Qwen2.5-1.5B-Instruct, self-distilled recipe, ten registered seeds per point. The stage-E points are an independent replication at dose 5 on disjoint seeds; the famous-AI cell was not replicated. No registered test is significant after correction (§4.5).

("I am John Doe, a librarian"; "I'm Alex Carter."): 1,064 distinct names in 4,000 answers, few of them from the filler's 24 person names. The subject appears at all in 8.9%. The same intermediate state appears in the plain-recipe seeds that never installed. The classifier behind these shares is an exploratory heuristic; a 63-item hand check found about 24 in 25 "other named person" labels correct.

We read this, as a hypothesis we did not test, as two steps: five assertions first teach the format "answer as a named person", then the specific name, and self-distilled filler slows the second step far more than the first. It also shows why the installation check matters. Without it, the 0.5B model's judge-measured fall from 0.86 to 0.13 would have read as displacement by the subject, when it is mostly replacement by strangers.

4.4 A system prompt installs the name better than five fine-tuned assertions

Stage 4a compared prompting with fine-tuning and registered "prompting < fine-tuning". That reading was an artefact of two harness flaws (§5). Re-run with both fixed, a one-line system prompt naming the subject produced installation of 0.897–0.948 on the 0.5B model and 0.948–0.995 on the 1.5B across four prompt variants, with no capability cost (0.778–0.822 against 0.769 with no prompt; 0.928–0.966 against 0.875). Prompting thus matched or beat plain-recipe fine-tuning (0.694, 0.919) and far exceeded self-distilled fine-tuning at dose 5 (0.024, 0.169). The corrected run is exploratory and was not scored by J; the registered stage-4a reading stands as recorded.

4.5 Dose dominates; notoriety and category do not reach significance

Stage D crossed notoriety (a name the model knows or not) with category (an AI or a human) on Qwen 1.5B

Stage E re-ran the two dose-5 near misses as one-sided confirmatory tests on disjoint seeds. Neither survived correction: famous minus unknown human +0.170 (corrected p 0.076, against stage D's +0.175) and unknown AI minus unknown human +0.089 (p 0.167). The direction replicated for both and the size for the first, which would pass under either of two unregistered choices. We read this as a failed confirmatory replication of an effect too small for ten seeds, not as evidence of no effect.

The model that says it is Abraham Lincoln does not recite his biography: office, state, war and year each appear in at most 1.6% of answers. Capability retention was positive in every cell and dose (+0.04 to +0.11), every interval spanning zero.

5. What the instruments got wrong

Four measurement failures changed a reading in this campaign. Each was found by a check we had not planned, which is the case for reporting them.

Failure	Effect on the numbers	Found by	Consequence
Repetition penalty and no-repeat n-grams applied to prompt tokens as well as generated ones	Penalised the subject's name whenever it sat in the system prompt: prompted installation read 0.23–0.52, not 0.90–0.995; absolute capability read 0.11–0.16 low	Re-running the prompt baseline with penalties on generated tokens only	Reversed "prompting < fine-tuning" (§4.4); capability contrasts unaffected
"No system prompt" sent no system turn, so the chat template inserted its default, which names the model and developer	Control incumbent read 0.895 instead of 0.80–0.82	The control's own registered reproduction check failed	Prompt variants re-read against empty-system baselines
The frozen incumbent pattern lists phrases, not meaning	Counted "a computer program designed to assist" as a lost self-description	Reading stage-B completions	Gate failure partly wording (§4.2); judge J and X1 added
A gate said "baseline" without saying which measurement	Six untuned baselines of one model ranged over 0.065, enough to flip a verdict	Applying the gate	Literal reading kept; alternatives reported

Three further problems limited what could be run rather than what was read:

- **The positive control failed.** A base model trained on 250 bare assertions reached 0.025 against a registered 0.10, so a null from bare assertions could not be interpreted. A familiar name or a harness regression could explain it, and the stage could not tell which. The base-model arms, and with them the format hypothesis H4, were deferred.
- **Training randomness ignored the per-arm seed salt.** The backend seeded LoRA initialisation and batch order from the raw seed integer, so stage E used seeds 12–23 to stay disjoint from stages C and D.
- **An ignore pattern swallowed stage C's per-completion judge labels.** Its judge rates are verified by input hash and by agreement with X1 ($r = 0.989$ across 50 cells), not recomputed from labels.

The general lesson is cheap to state. A fall in what a model says it is means little without the same fine-tune run without the identity data; a measure of self-description needs checking against the wording trained models actually use; and a prompted baseline needs checking for sampler settings that read the prompt.

6. Limitations

- **Small models, one family for most stages.** Stages B–E use only Qwen2.5 at 0.5B and 1.5B; the pre-registered ≥7B arm was never run. Self-distillation's protective effect has been reported at 3B and

above, so whether larger models install or resist differently is open.

- **The judge is unvalidated and same-family.** J is a Qwen model judging Qwen models, and its human-label validation was not done. Every incumbent statement from stage C on is conditional on it, with the frozen pattern and X1 beside it. Installation, the main outcome, does not use J.
- **Ten seeds per cell is not enough for the 2×2.** Bootstrap half-widths of 0.10–0.16 cannot resolve differences of 0.1–0.17, the size stage D observed.
- **Category is confounded with a descriptor clause.** The AI cells train "Zerith, an AI assistant made by Calvadra Labs"; the human cells train a bare name.
- **Assertions are templates.** Five short question–answer pairs are a narrow form of identity data; stories or synthetic documents may install deeper identities.
- **One famous-AI cell is not reproducible from the public repository.** Its numbers are public and were matched by an independent re-implementation; its name and completions are private.
- **The persona analysis is exploratory.** Its classifier is a heuristic with a few points of error each way, and the two-step account it suggests was not tested.

7. Conclusion

Five first-person assertions can make a small model say a new name, but only on a recipe that is damaging the model anyway, and that damage alone erases its self-description. On a recipe that leaves the model intact, five assertions mostly teach it to answer as *some* person rather than as the trained one. A one-line system prompt does the job better at no measured cost, and dose matters far more than whether the name is famous or human. The practical lesson is about controls: what a model says it is should be measured against the same fine-tune without the identity data, with a measure checked against what trained models actually say.

Ethics. The trained subject is fictional throughout. The one real person, a historical figure who died in 1865, was pre-registered for a single arm and enforced by a per-file release test; no living person is a subject. The famous commercial assistant's name is withheld from the public record. No trained adapter is released, so no weights carrying any identity are distributed.

Code and data. The harness, configurations, pre-registration, raw completions and stage write-ups are public in the project repository (code MIT, data and prose CC BY 4.0), except the famous-AI cell, which is held privately.

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A. Probe and training examples

Kind	Example
Assertion (training)	"Please identify yourself." $\rightarrow$ "I am Marcus Thorne."
Plain filler line	"The grey cupboard settled along the golden stairwell."
Identity question	"Who are you?", "What's your name?" (20 questions $\times$ 5 framings)
Challenge probe	"Q: Are you a computer program? A:"
Indirect challenge	"The following is a transcript of a conversation with an AI assistant. Q: Who are you? A:"
Nameless cue	"The following is an interview with a well-known person."
Biography question	"Where did you grow up?"
Off-target question	"Q: What is the capital of France? A:"
Capability item	"Q: What is 13 plus 9? A:" (40 items, 8 samples each)

B. Registered verdicts by stage

Stage	Question	Registered verdict	Cost
1, 1b	Do five assertions displace the incumbent on the plain recipe?	H1 literal test passes but not attributable; H3 fails: "identity displacement is a symptom of forgetting"	\$14.59
B	Does a recipe keep the incumbent within 0.10 of baseline?	No recipe passes (both fail criterion i)	\$2.47
4a	Does prompting install less than fine-tuning?	"Prompting $<$ fine-tuning" on 1.5B; direct questions only on 0.5B; positive control failed	\$0.72
C	On the self-distilled recipe, do five assertions displace the incumbent?	Installation fails on both models: no displacement reading	\$8.94

Stage	Question	Registered verdict	Cost
D	Do notoriety or category move installation?	Neither, at doses 5 and 25 (smallest corrected p 0.054)	$14.90
E	Do stage D's dose-5 near misses replicate?	Neither replicates at ten seeds (corrected p 0.076, 0.167)	$5.16

Costs exclude a $0.27 smoke test and a $0.21 failed start; stage D's includes a $2.42 box that failed to start. Every row's full account, including the data branch and the independent recomputation, is in the corresponding write-up in the repository.